

JC4003: Natural Language Processing - Group Project Assessment

The Architect's Blueprint: Designing an LLM-Based Academic Paper Rating System

Issue Date: October 14, 2025

Submission Deadline: December 5, 2025, 22:00 SCNU Local Time

Weighting: 25% of Final Grade

Submission Format: A single PDF document submitted via Blackboard.

1. Introduction and Learning Objectives

Welcome to your final year NLP group project. The goal of this assessment is not to build a functional system from scratch but to engage in the critical design and planning process that precedes any successful AI project. You will work in groups to produce a detailed design document for an innovative system that uses a Large Language Model (LLM) to automate the rating and feedback process for academic papers.

This project will challenge you to:

- Apply your understanding of modern NLP concepts, particularly those related to LLMs.
- Think critically about system architecture, data requirements, and model selection.
- Develop robust evaluation strategies to measure system performance.
- Consider the novelty of your ideas and the ethical implications of AI systems.
- Enhance your technical writing and collaborative skills.

2. Project Overview

Peer review is the cornerstone of academic progress, yet it is often a time-consuming and subjective process. Your task is to design a system that assists in this process by automatically rating academic papers based on a set of predefined criteria.

The proposed system should be able to:

- Accept a full-text academic paper (e.g., in PDF or plain text format).
- Analyze the paper against a multi-faceted rubric (which you will define).
- Generate a numerical score for each criterion.

- Provide constructive, text-based feedback to justify the ratings.

Your group's final output will be a comprehensive **design report** that details your proposed solution, justifies your design choices, and outlines a clear plan for evaluation.

3. Required Sections of the Design Report

Your report should be structured professionally, similar to a grant proposal or a technical report. It must include the following sections:

Part A: The Vision (15%)

1.0 Abstract/Executive Summary (approx. 250 words): A concise summary of the entire project. It should briefly describe the problem, your proposed solution, its key features, the methodology, and its potential impact.

2.0 Introduction & Motivation (approx. 1 page): • **Problem Statement:** Clearly define the challenges and limitations of the current academic peer-review process.

- **Project Goals:** State the primary objectives of your proposed system. What do you aim to achieve?
- **Significance:** Explain why this problem is important to solve and who the target users would be (e.g., conference organizers, journal editors, students).

Part B: The Foundation (25%)

3.0 Literature Review (approx. 2-3 pages): • Survey existing work in automated essay/paper scoring, text summarization, and feedback generation.

- Discuss the evolution from traditional NLP methods to modern LLM-based approaches for these tasks.
- Identify gaps in the current research that your proposed system aims to address.

Part C: The Blueprint (30%)

4.0 System Architecture and Design (approx. 4-5 pages): This is the core of your project.

- **4.1 Overall Architecture:** Provide a high-level diagram and description of the system's workflow, from input to output.
- **4.2 Data Pipeline:**
 - **Data Source:** ACL Conference: Summarization Track (<https://aclanthology.org/events/acl-2025/>) [Note: replace 2025 with 2024, 2023, etc to find previous ACL publications]
 - **Data Requirements:** What kind of data is needed to train (if applicable) and evaluate your system? (e.g., datasets of papers with scores, reviewer comments).
 - **Data Pre-processing:** How would you handle different input formats (e.g., PDF parsing), extract relevant sections (abstract, methodology, conclusion), and clean the text?
- **4.3 Core Model(s):**

- **LLM Selection:** Which specific LLM(s) would you use (e.g., GPT-4, Gemini, Llama 3, a fine-tuned open-source model)? Justify your choice based on factors like performance on academic text, context window size, cost, API availability, and ethical considerations.
- **Interaction Strategy:** Explain whether you would use the model via zero-shot prompting, few-shot prompting, or a fine-tuning approach. Justify this critical decision.
- **4.4 Prompt Engineering / Fine-Tuning Details:**
 - **Rating Rubric:** Define the specific criteria your system will use to rate a paper. Examples include **Clarity, Methodology, Novelty, Contribution to the Field, and Reproducibility**. Justify your choice of criteria.
 - **Prompt Design:** Provide detailed examples of the prompts you would use to instruct the LLM to rate the paper against your rubric. How would you structure the prompt to ensure reliable and structured output (e.g., requesting JSON)?
- **4.5 Output and User Interface:**
 - Describe the format of the final output. Will it be a summary report, an annotated rubric, or something else?
 - Include mock-ups or sketches of how the results would be presented to the end-user.

Part D: The Validation (20%)

- 5.0 Novelty and Contribution (approx. 1 page):** • Clearly articulate what makes your design innovative. Is it the choice of model, the prompting strategy, the rating rubric, the user interface, or a combination of factors?
- How does your proposed system improve upon the existing solutions discussed in your literature review?
- 6.0 Evaluation Plan (approx. 2 pages):** • **Metrics:** Define the specific quantitative metrics you would use to evaluate your system's performance. Examples include:
- **Inter-rater Reliability:** Cohen's Kappa or Krippendorff's Alpha to compare system ratings with human expert ratings.
 - **Correlation:** Pearson or Spearman correlation between system scores and human scores.
 - **Accuracy/MSE:** Mean Squared Error for numerical scores.
 - **Qualitative Analysis:** A plan for human assessment of the generated feedback's quality (e.g., relevance, constructiveness, clarity).
 - **Experimental Setup:** Describe the experiment you would run. How many papers? How many human raters? How would you ensure a fair comparison?

Part E: The Reality Check (10%)

- 7.0 Ethical Considerations and Limitations (approx. 1 page):** • Discuss potential biases (e.g., against certain research areas, writing styles, or non-native English speakers) that the LLM might exhibit and propose strategies to mitigate them.
- Address concerns related to data privacy, plagiarism, and the potential over-reliance on automated systems.

- Acknowledge the limitations of your proposed system.

8.0 Project Timeline: • Provide a Gantt chart or a detailed timeline illustrating the hypothetical project plan for one semester, from initial research to final evaluation.

4. Group Work and Collaboration

This is a group project. Groups should consist of **4-6** students. Cross-programme group is **NOT** allowed. You are expected to collaborate on all aspects of the design document. At the end of your document, please include a "Statement of Contribution" where each group member briefly outlines their primary contributions to the project.

5. Marking Rubric

Section	Criteria	Weight
Part A: The Vision	Clarity of problem statement, motivation, and project goals. Professionalism of the abstract.	15%
Part B: The Foundation	Comprehensiveness and relevance of the literature review. Clear identification of research gaps.	25%
Part C: The Blueprint	Detail and justification of the system architecture, data pipeline, model selection, and prompt design. Coherence of the overall design.	30%
Part D: The Validation	Clear articulation of novelty. Rigor and appropriateness of the proposed evaluation plan and metrics.	20%
Part E: The Reality Check	Thoughtful discussion of ethical issues and limitations. Realistic project timeline.	10%
Overall Quality	Clarity of writing, professional formatting, logical flow, and adherence to submission guidelines.	(100%)

Good luck, and we look forward to seeing your innovative designs!